FET small-signal II

Cheat sheet · Day 20 · Thu 12 Nov · pack E12

Reference-sheet lines

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FET SMALL-SIGNAL II (Week 8 second half)
gm: JFET/D-MOS gm0 = 2IDSS/|VP|, gm = gm0(1 - VGSQ/VP)
    E-MOS gm = 2k(VGSQ - VGS(Th)),  k = ID(on)/(VGS(on)-VGS(Th))^2
    Any:  gm = 2ID/|VGS - Voff|      rd = 1/gos  (not given -> infinity)
Sedra: Vov = VGS - Vt, ID = 1/2 unCox (W/L) Vov^2, gm = unCox(W/L)Vov = 2ID/Vov
    course k = 1/2 unCox (W/L).  If W/L is asked, given "k" = unCox.
CONFIG        Zi                        Zo               Av
CS (byp/VD)   RG or R1||R2              RD||rd           -gm(RD||rd)
CS unbyp RS   RG                        ~RD              -gmRD/(1+gmRS+(RD+RS)/rd)
Src follower  RG                        rd||RS||1/gm     gm(rd||RS)/(1+gm(rd||RS))  <1
Common gate   RS||(rd+RD)/(1+gm rd)     RD||rd           (gmRD+RD/rd)/(1+RD/rd) ~ gmRD
              ~ RS||1/gm
Drain fdbk    (RF+rd||RD)/(1+gm(rd||RD)) RF||rd||RD     -gm(RF||rd||RD) ~ -gmRD
              ~ RF/(1+gmRD)
LOADING: put RL in parallel with output resistor (RD, or RS for follower).
         Avs = Zi/(Zi+Rsig) * AvL   (big effect only for CG)
CASCADE: AvT = Av1*Av2, Zi = Zi1, Zo = Zo(last), RL1 = Zi2; Vo = AvT*Vi.
         Don't round stage gains before multiplying.
DESIGN:  RD||rd = |Av|/gm -> RD = 1/(gm/|Av| - 1/rd); standard value; recheck
         VDS > VGS - VP (saturation).
Sedra quick: CG matched -> 1/gm = Rsig, Gv = 1/2 gmRD; follower Ro = 1/gm,
         Av = RL/(RL+1/gm); ID = gm Vov/2; W/L = gm/(unCox Vov).
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